Sepehr Mahmoudian

SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI



EMAIL sepmhn@gmail.com

GITHUB github.com/sepahead

LINKEDIN linkedin.com/in/sepehr-mahmoudian

Berlin-Brandenburg • available from Jan 2027 • German citizen • EN fluent / DE good / Persian native

PROFILE

Builds agent harnesses for domain-specific applications, knowledge graphs and Graph RAG; orchestrates agentic workflows and evaluates LLMs, VLMs, VLAs and world models. Expertise in formal methods and verification.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction

CONTEXT ENGINEERING • DATA PROVENANCE • MODEL EVALUATION • REINFORCEMENT LEARNING

AI ENGINEERING

10+ YEARS

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

PROFESSIONAL EXPERIENCE

Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

Self-employed - PhD candidate at Technische Universität Darmstadt Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), Vision-Language-Action systems, VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

Senior AI Engineer

2026.01 - 2026.03

Neural Frames UG Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python evaluation and inference pipelines; applied formal methods and verification

AI Engineering Resident

2025.09 - 2026.01

Merantix AI Campus (Hacker Room) Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

Senior AI Engineer

2024.07 - 2025.07

Vision Labs UG Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

Research Associate

2021.07 - 2024.07

Freie Universität Berlin Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

Research Associate
2019.07 - 2020.06

Georg-August-Universität Göttingen Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

Research Associate
2017.03 - 2019.07

Goethe-Universität Frankfurt Frankfurt am Main, Germany · Simulation and statistical analysis of neural-network models

Visiting Scientist
2017.03 - 2019.03

Max Planck Institute for Dynamics and Self-Organization Göttingen, Germany · Statistical methods for neural-network analysis

Research Associate
2015.02 - 2017.03

Forschungszentrum Jülich Jülich, Germany · Neural-network simulations and reinforcement-learning experiments

Software Developer
2010.09 - 2012.03

Centre for Content Creation Cyberjaya, Malaysia · Award-winning iPhone/iPad applications and digital products

EDUCATION

PhD Candidate (Dr. rer. nat.)
2020.04 - Present

Technische Universität Darmstadt Darmstadt, Germany

Thesis: Partial Information Decomposition and Closed-Loop Simulation of Neural and Embodied Systems on Formally Verified Foundations

- New mathematical PID analyses and information-theoretic representation analysis; formal verification of selected results and protocol properties with Lean and SMT
- Engram's domain-specific AI agent produces neural-model candidates for closed loop neural network simulations in reproducible computational-neuroscience workflows, with provenance and bounded compilation and simulation checks

Master's by Research in Biosciences: Neuroscience
2012 - 2014

University College London London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

Bachelor of Science (Honours) in Computer Science
2009 - 2012

Anglia Ruskin University Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

SELECTED AWARDS & FUNDING

EDTH / 2025
Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub

HUMAN BRAIN PROJECT
Research funding

NEAR HORIZON
Confidential-computing grant

UCL & SHEFFIELD
Competitive academic scholarships

IOS APPLICATIONS
15 international product awards



SIGNATURE
Sepehr Mahmoudian

PLACE Berlin, Germany
DATE 13 November 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects
github.com/sepahead

HUGGING FACE Models & datasets
huggingface.co/torusprime

SELECTED PROJECTS

engram

★ 15 · RESEARCH PROTOTYPE

PROVENANCE-CONSTRAINED NEURAL R&D

Domain-specific AI agent turns qualified evidence into neural-model candidates, with bounded compilation and simulation checks.

Neural Networks · Knowledge Graphs · Closed-loop Sim

pid-rs

★ 15 · V0.9 SOURCE PRERELEASE

VERIFIABLE INFORMATION DECOMPOSITION

Separates redundant, unique and synergistic information, with selected Lean/SMT proofs and declared estimation assumptions.

Rust / Python · Information Theory · Lean / SMT

prisoma

★ 5 · V0.9 PRERELEASE / ACTIVE R&D

AUDITABLE CLOSED-LOOP EXPERIMENTS

Simulation environment for closed-loop VLA and world-model experiments, with capture, replay and provenance.

Rust / Python · VLA + World Models · Embodied Sim

NCP

★ 16 · SDK 1.0.1 / NCP 2.0 PREPARED

NEURAL CONTROL PROTOCOL

Couples separately run simulators, robots and tools: no step runs twice, failures have explicit outcomes, sessions are verifiable hash chains.

Rust / Python · Lean Proofs · Fault Injection

manwe

★ 6 · ALPHA

AIRSPACE PERCEPTION

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python · PyTorch · Multimodal Perception

crebain

★ 23 · RESEARCH PROTOTYPE

SPATIAL SITUATIONAL AWARENESS

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust · Sensor Fusion · Native Inference

melkor

★ 13 · HARDENING

GAUSSIAN-SPLAT INTEGRITY

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++ · Scene Reconstruction · Asset Validation

cobot-atlas
relief-atlas

★ 8 · ↓ 3,271 · PUBLISHED + DOI

AI-GENERATED SIMULATION DATASETS

Cobot Atlas publishes robot-simulation meshes with a DOI; Relief Atlas develops disaster-relief assets and generation pipelines.

Python · Hugging Face · 3D / VLA

galadriel

★ 8 · RESEARCH PROTOTYPE

CROSS-SENSOR CONSISTENCY

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust · NIS / CUSUM · Signed Correlation

cortexel

★ 5 · 0.10.0-DEV.0

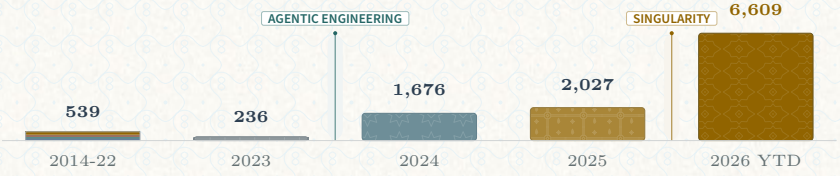
AUDITABLE NEURAL FIGURES

Checks scientific figures against declared contracts before rendering, so charts provably match their data and stay reproducible.

Figure Contracts · Deterministic SVG · Provenance

PORTFOLIO & ENGINEERING ACTIVITY

11,087	23	125	203
GITHUB CONTRIBUTIONS SINCE 2014	NON-FORK PUBLIC REPOSITORIES	STARS ON PUBLIC REPOSITORIES	ACTIVE DAYS IN 2026



CURRENT MOMENTUM

3.26× 2026 YTD contribution volume versus the full 2025 calendar year

6,609 YTD • 2,027 in 2025

CONTRIBUTION BY WEEKDAY



6,609 contributions • 2026 YTD

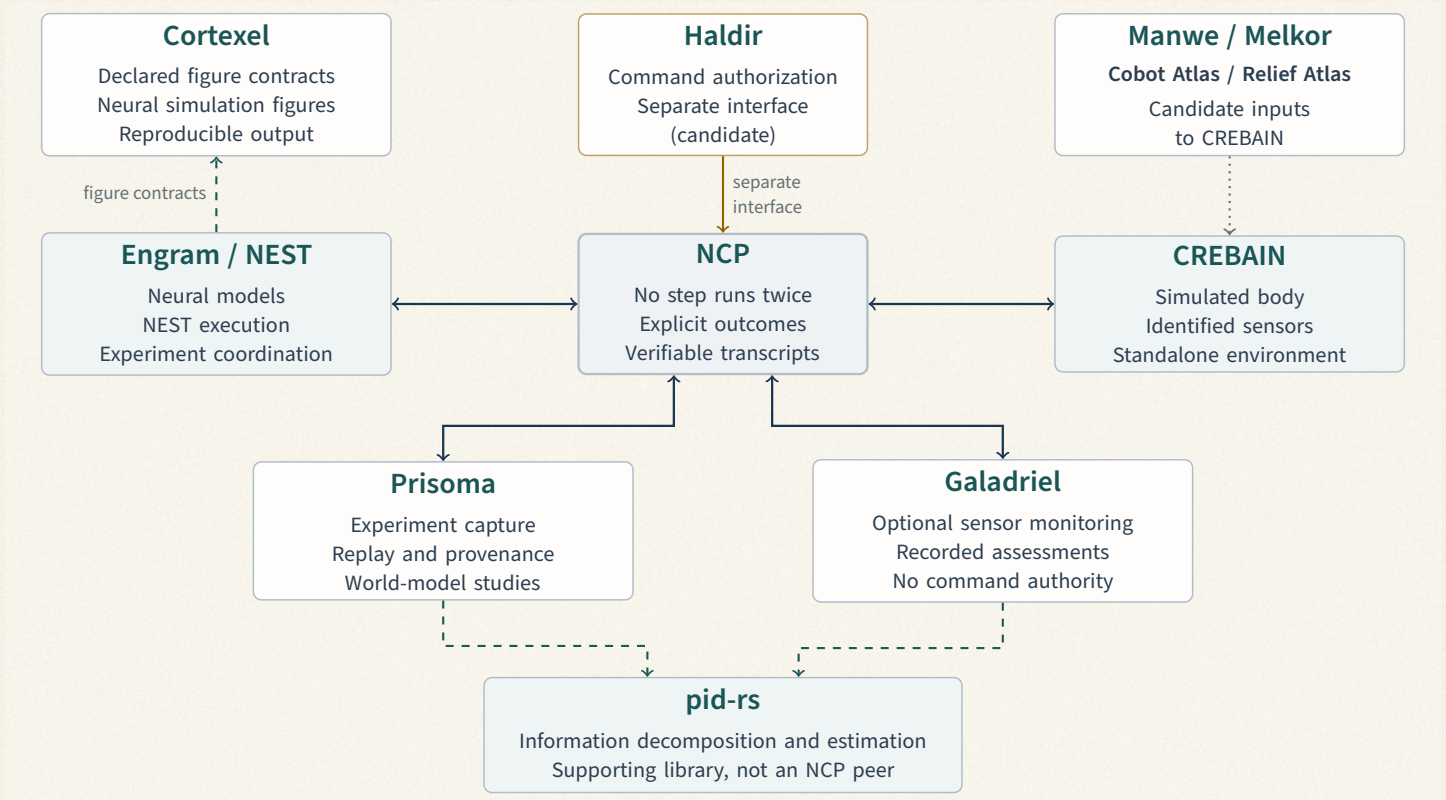
Mon 17.2%	Tue 13.5%
Wed 12.3%	Thu 9.8%
Fri 11.0%	Sat 18.9%
Sun 17.4%	

TECHNICAL SKILLS

AGENTIC AI	LLMs / VLMs · Agent Harnesses · AI Agents · Tool Use · Model Evaluation · Inference Pipelines
MACHINE LEARNING	Python · PyTorch · Agentic Vision · Multimodal AI · Vision-Language-Action / VLA Evaluation
KNOWLEDGE & RETRIEVAL	Knowledge Graphs · Graph RAG · RAG · Embeddings · Vector Search · Reranking · Data Provenance
SYSTEMS & MLOPS	Rust · C++ · FastAPI · ONNX · TensorRT · CUDA · Metal / Core ML · Docker
CLOUD & INFRASTRUCTURE	GCP · AWS · Azure · Kubernetes · Terraform · GitHub Actions · Linux · Zenoh
FORMAL VERIFICATION	Formal Methods · Lean · SMT / Z3 · Model Checking · Differential Testing

RESEARCH ECOSYSTEM

Independently runnable projects for neural and embodied experiments. Each program owns its state; NCP supplies shared contracts rather than a central runtime. The map distinguishes interfaces present in source from candidate inputs.



Local requests and results in source
Uses a supporting library
Candidate input; adapter not established

Neural loop

Engram owns neural execution in NEST. CREBAIN owns the simulated body and sensor state. Over NCP 2, spiking controllers in NEST have flown one or several simulated drones, each action traced to the observation it answered; the body keeps final actuation authority.

Evidence and analysis

Prisoma captures experiment exchanges. Galadriel records optional diagnostics. Their source analysis paths use pid-rs; Engram's figure path uses Cortexel. Capture and monitoring grant no control authority.

Qualification

Current examples start from explicit plans and scenes. Integration qualification remains partial. Manwe, Melkor and the atlases are candidate perception or asset sources. Arbitrary-paper reproduction remains a research goal.

Why NCP

Reasoning

Separately owned programs fail independently: a reply can be lost, a step can run twice, and a stale command can reach a newer state.

Features

One round trip per step; a repeated request never runs twice; every fault has an explicit outcome; SHA-256 digests chain each session into a verifiable transcript; owner rules proved in Lean.

Justification

MUSIC needs one MPI job and, in the tested NEST build, delivered exact spikes only with one thread. gRPC, ZeroMQ, Zenoh and HTTP build in none of these guarantees.

Performance

9 µs per call in Rust and 39 µs in Python; 23 to 32 µs added to each 1 ms NEST step (NCP 1: 228 to 248 µs); no wrong result in 8,730 fault trials.